

Speech Signal Processing: Sampling, Quantization, and Encoding

Lab 1 Task Report

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Report

1. Introduction

This report presents the process and results of processing a speech signal using sampling, quantization, and encoding techniques. The speech signal undergoes the following stages:

- **Sampling** – Reducing the original signal's sampling rate.
- **Quantization** – Discretizing the signal amplitude to a set of finite levels.
- **Encoding** – Mapping quantized values to binary representations for digital storage and transmission.

The following sections provide detailed observations and discussions on each stage of the process.

2. Effect of Sampling

The original speech signal is sampled at a higher rate (typically 44.1 kHz, as in the harvard.wav file). In the experiment, the sampling rate was reduced to 8 kHz for comparison. The original and the sampled signals are presented below:

Original Speech Signal (Higher Sampling Rate):

Sampling Rate (F_s): 44.1 kHz (assumed).

Signal Characteristics: The higher sampling rate preserves more details of the speech signal, capturing both low and high-frequency components.

Sampled Speech Signal (Lower Sampling Rate, 8 kHz):

Sampling Rate (F_{s_new}): 8 kHz.

Signal Characteristics: The signal is resampled to 8 kHz. Reducing the sampling rate causes a loss of higher frequency components that are not captured in the lower sampling range. High-pitched sounds and fast changes in the waveform may be aliased or lost. This results in a lower fidelity representation of the original signal.

Discussion:

As seen from the comparison between the original and resampled signals, reducing the sampling rate causes noticeable degradation in signal quality. At 8 kHz, high-frequency details of the speech signal, such as sharp transitions, are not accurately captured. This phenomenon is referred to as aliasing. Higher sampling rates are preferable when preserving signal quality is important, but lower rates are often used in speech communication to reduce data size, as the human ear is less sensitive to higher frequencies (above 8 kHz).

3. Effect of Quantization

Quantization involves mapping the continuous amplitude values of the signal to discrete levels. This process introduces a form of distortion known as quantization error, where the signal amplitude is rounded to the nearest available quantization level. The quantization process for the resampled speech signal is performed with an 8-bit depth.

Original vs Quantized Signal:

Original Signal: A continuous range of amplitude values.

Quantized Signal (8-bit): The signal is quantized to 256 discrete levels (since $2^8 = 256$).

Discussion:

The quantized signal differs from the original one in terms of the smoothness of the waveform. The quantized signal has stepped values, leading to a loss in accuracy when representing the true amplitude of the original signal. Increasing the number of bits in quantization improves the signal's accuracy, but also increases the number of levels, leading to a larger file size.

With 8-bit quantization, the signal quality is somewhat reduced, but for typical speech signals, 8 bits may be sufficient for maintaining intelligibility while reducing data size. Using higher bits (e.g., 16-bit) would improve quality but at the cost of more data.

The effect of quantization can be seen as a balance between signal quality and data size. While increasing the number of bits reduces quantization noise, it also increases the file size. In practical systems, quantization depth is often chosen based on the required signal quality and the available storage or transmission bandwidth.

4. Encoding Process

The encoding process maps the quantized values to binary representations. The resulting binary sequence allows the signal to be stored and transmitted in a digital format. In this experiment, the quantized signal was mapped to an 8-bit binary format.

Binary Representation of Quantized Signal:

Each quantized sample is converted into its binary equivalent, represented as 8-bit binary values. The encoding process ensures that the signal can be efficiently transmitted or stored in a digital medium.

Discussion:

The encoding process allows for precise representation and manipulation of the signal in digital communication systems. The binary encoding helps in eliminating errors during transmission and ensures that the signal can be reconstructed at the receiver end.

Binary encoding is a critical component of modern digital communications, where signals need to be transmitted over networks or stored in devices. This step ensures that each quantized sample is represented in a form that is suitable for digital systems, such as audio files or communication protocols.

5. Conclusion

This experiment illustrates key concepts in speech signal processing, particularly in digital signal representation through sampling, quantization, and encoding. The following conclusions can be drawn:

- **Sampling:** Lowering the sampling rate results in a loss of high-frequency information, which may degrade the quality of the signal.
- **Quantization:** The number of bits used for quantization directly impacts the signal quality. Higher bit depths improve the fidelity of the signal but increase the data size.
- **Encoding:** The encoding step is crucial for digital signal representation, allowing for efficient storage and transmission of quantized signals.

By understanding the effects of these stages, one can make informed decisions about how to balance signal quality and data size in practical systems, such as audio compression and digital communication.

MATLAB Code

```
1 close all
2 clear
3 clc
4
5 [x, Fs] = audioread('harvard.wav'); % Replace with your audio file
6
7 if size(x,2) > 1
8     x = mean(x,2);
9 end
10
11 % Plot the original signal
12 figure(1);
13 subplot(3,1,1);
14 plot(x);
15 title('Original Speech Signal');
16 xlabel('Samples');
17 ylabel('Amplitude');
18
19 % Sampling
20 Fs_new = 8000; % New sampling frequency (8 kHz)
21 y = resample(x, Fs_new, Fs);
22
23 subplot(3,1,2);
24 plot(y);
25 title('Sampled Speech Signal (8 kHz)');
26 xlabel('Samples');
27 ylabel('Amplitude');
28
29 % Quantization
30 nbits = 8; % Number of bits for quantization
31 L = 2^nbits; % Number of quantization levels
32 x_min = min(y);
33 x_max = max(y);
34 delta = (x_max - x_min) / (L-1);
35 quantized_signal = round((y - x_min) / delta) * delta + x_min;
36
37 subplot(3,1,3);
38 plot(quantized_signal);
39 title('Quantized Speech Signal (8-bit)');
40 xlabel('Samples');
41 ylabel('Amplitude');
42
43 % Encoding
44 binary_encoded = dec2bin(round((quantized_signal - x_min) / delta),
45     nbits);
46
47 disp('Binary Encoded Samples:');
48 disp(binary_encoded(1:10,:));
49
50 % Save the quantized audio
51 audiowrite('quantized_speech.wav', quantized_signal, Fs_new);
```

Output

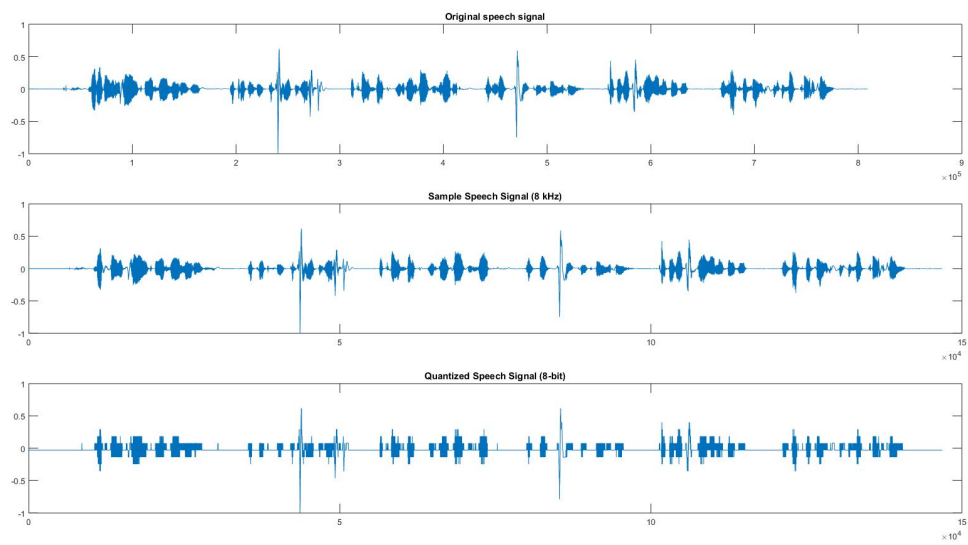


Figure 1: Plot showing original, sampled, and quantized speech signals